

Predictive Maintenance for Industrial Machines

Machine Learning 1, Final Project

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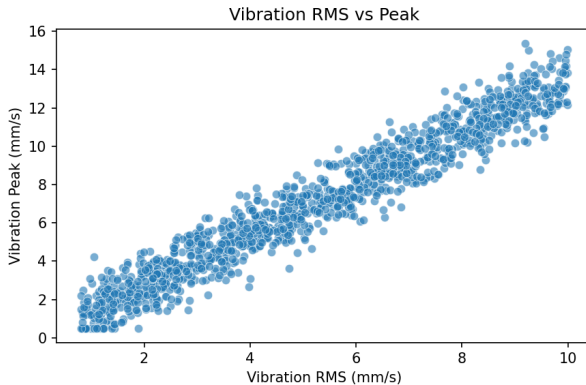
Objectives

- Estimate, for each FactoryPulse machine, the probability of failure within the next 30 days
- Convert that score into a concrete maintenance policy: which machines get a preventive check, and at what cost in false alarms

Initial Exploratory Insights

- Machine failures only occur in a small proportion of the data (13.4%).
- Features regarding machine age, operating hours, vibration measures and prior failures are the most impactful on failure.
- The class imbalance shapes everything that follows: metrics, model training, and the decision threshold.

Vibration Analysis



The two vibration features are strongly related, which may introduce some redundancy.

Constant Predictor as a Baseline

- The baseline always predicts the majority class, i.e., non-failure of the machine.
- High accuracy observed (0.866), but not representative because of class imbalance.
- Balanced accuracy of 0.5, indicating no ability to distinguish failures from non-failures.
- PR-AUC of 0.134, exactly the failure rate: the score of a model with no information.
- Conclusion: accuracy is out; PR-AUC is our primary metric.

Evaluation Protocol

- All comparisons use stratified 5-fold cross-validation on the training file, preserving the 13.4% failure rate in every fold.
- Scaling lives inside a pipeline, fit on each fold's training part only: no data leakage.
- Hyperparameters are selected by cross-validation, never on the test set.
- The test file is opened exactly once, after the model and the threshold are frozen.

Candidate Models

- Logistic regression: plain, with balanced class weights, and with a tuned L2 penalty
- Linear discriminant analysis
- Gaussian Naive Bayes
- K-nearest neighbors, with k tuned by cross-validation

All methods from the course material; each makes a different assumption about how features relate to failure.

K-Nearest Neighbors



img/knn_k_selection.png

- Scaling is mandatory: distances would otherwise be

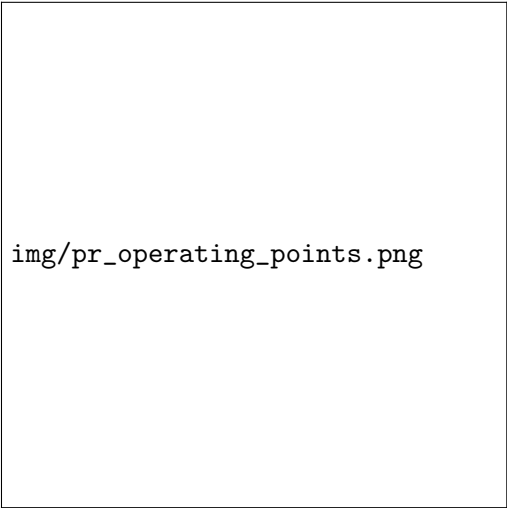
Logistic Regression

- Penalty strength tuned by cross-validation ($C = 0.1$), the same selection problem as choosing λ for Ridge.
- Balanced class weights compensate the imbalance during training.
- Best model in the comparison: CV PR-AUC 0.450 ± 0.110 .
- Readable coefficients: prior failures, vibration, operating hours and machine age increase risk; operator experience decreases it.

Choice of Final Model

- CV PR-AUC: logistic regression 0.450, LDA 0.438, kNN 0.435, Naive Bayes 0.430.
- Fold-to-fold standard deviation is 0.110: the top models overlap, so we cannot claim a statistically clear winner.
- With overlapping scores, we prefer the simpler, interpretable model: logistic regression, whose coefficients the reliability team can read.
- Naive Bayes has the highest ROC-AUC, but ROC-AUC is inflated by the large negative class; PR-AUC matches the task.

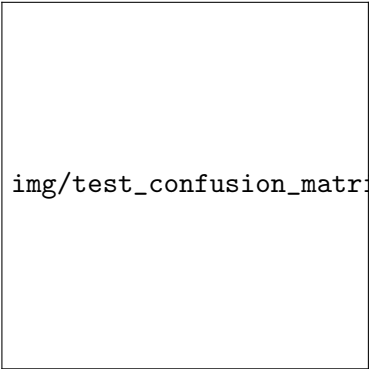
A Preventive Policy



`img/pr_operating_points.png`

- The threshold is the policy. We price six recall guarantees on out-of-fold predictions.

Final Evaluation Results



`img/test_confusion_matrix.png`

- Test set used once, threshold frozen: 33 of 37 failures caught (recall 0.89), 47% of the fleet flagged, precision 0.20.
- PR-AUC 0.40 against a 10.6% base rate: ranking almost four times better than random.
- Precision dip vs training folds is base-rate arithmetic (10.6% vs 13.4% failures), not model decay; ROC-AUC held at 0.82.

Limitations and What the Model Misses

- The missed failures look like healthy machines: near-healthy vibration and no failure over the last year. A limit of the sensors, not of the threshold.
- Snapshot data: the model cannot see trends such as vibration rising over weeks.
- Failure and maintenance costs are not in the data; the operating-point table lets FactoryPulse move the recall floor the day real costs exist, without retraining.
- The threshold assumes a stable failure rate: monitor the flag rate in production and re-pick from the table if it drifts.